



King Abdulaziz University

Faculty of Computing & Info Tech.

Computer Science Department

Final Lab Exam (2023)

Course: CPCS203 (Programming II), section:

Term: **Winter** Semester (2023), Time: **100 Minutes**

Name :

ID :

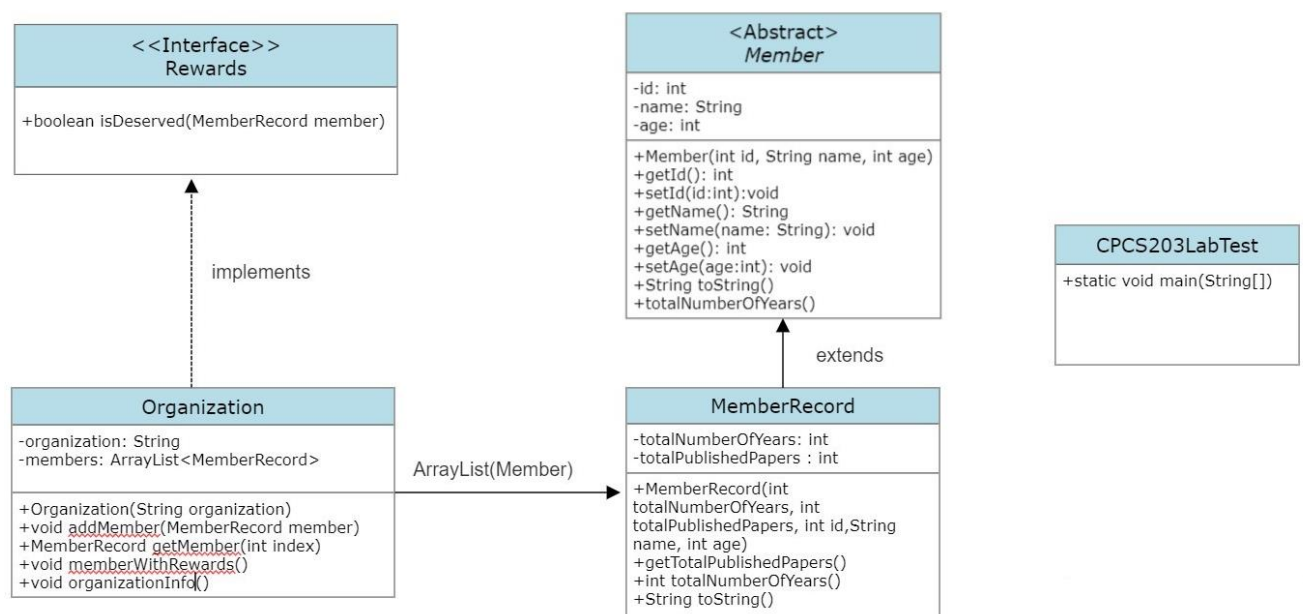
General Instruction:

- (1) Project Name must be ID_Section, for example 221908489_F3
- (2) Upload the project folder in the BLACKBOARD by compressing it as a zip file, ex: 221908489_F3.zip,
Go to documents\NetBeansProjects\YouProjectFolder – select and right click – pick - send to – compressed folder. Now, upload the compressed folder.
- (3) How to upload the solution file? It is the same, as, how you uploaded the assignments.
- (4) Make sure you have uploaded the correct folder. YOU WILL GET **ZERO** IF THE FILE IS NOT UPLOADED CORRECTLY OR CANNOT BE OPENED.
- (5) **Imp!!!!!!! Plagiarism will be checked by the software, and above 80% matching will result in scoring ZERO in the lab exam.**

Introduction:

You will make a simple and small program for a scientific organization like IEEE that rewards its members. Every organization, like the IEEE, has records for its members. For example, every member has an ID, name, age, totalNumberOfYears, and totalPublishedPapers with it.

You have to create the following **classes and interface** with their attributes (data fields) and methods (actions), as shown in the following **UML diagram**:



Zoom to enlarge!

Description:

1- “**Member**” is an **abstract superclass**. As shown in the above UML class diagram, it has an abstract method **totalNumberOfYears()** which returns an **integer** value [Refer to above UML diagram]. **[1 mark]**

2- ‘**MemberRecord**’ is a **sub-class** of Member [Refer to above UML diagram]. **[1 mark]**

The method you will override must return **totalNumberOfYears** if it is greater than zero; otherwise, -1.

3- ‘**Rewards**’ is an **interface** with an **abstract method (isDeserved)** [Refer to above UML diagram] **[1 mark]**

4- ‘Organization’ is a concrete class that implements ‘Rewards’ interface. This class must use ArrayList to store MemberRecord objects. Methods details are as follows: [Refer to above UML diagram] [2 marks]

4(a) “isDeserved()” method returns true or false based on the following rule: if a member has more than 8 totalNumberOfYears AND totalPublishedPapers => 4 (Greater than or equal to 4) returns true otherwise false. [2 marks]

Hint: check totalNumberOfYears and totalPublishedPapers.

4(b) “memberWithRewards()” method displays member details along with the Reward for those who qualify for it. [2 marks]

Hint: You will use 4(a) to get the 4(b) result, as shown below.

```
-----
-----List All Members with Reward-----
-----
Member {ID=1001, name=Ahmed, age=31}
The Record {total Numbers of Years=8, total Published Papers =6}
Reward: Discount 50%
-----
Member {ID=1002, name=Khalid, age=40}
The Record {total Numbers of Years=10, total Published Papers =9}
Reward: Discount 50%
-----
```

4(c) “organizationInfo ()” method display All member details in the organization as shown below. [2 marks]

```
Organization Name: IEEE
-----
-----Member of the organization-----
-----
Member {ID=1001, name=Ahmed, age=31}
The Record {total Numbers of Years=8, total Published Papers =6}
-----
Member {ID=1002, name=Khalid, age=40}
The Record {total Numbers of Years=10, total Published Papers =9}
-----
Member {ID=1003, name=Ibrahim, age=41}
The Record {total Numbers of Years=4, total Published Papers=3}
-----
Member {ID=1004, name= Mohmmmed, age=45}
The Record {total Numbers of Years=5, total Published Papers=5}
-----
Member {ID=1005, name= Sami, age=50}
The Record {total Numbers of Years=7, total Published Papers=2}
```

5- Test Your Solution ((CPCS203LabTest class (the main class):

After creating and implementing the classes and the interface as shown in the above UML diagram, you must create the main class, which has the main method, to test your solution: **[4 marks]**

- 1) Create Objects as required to run the program.
- 2) Let the user enter all the data, so you should use the Scanner class.
- 3) The output should be the same as the following:

A Sample Output

run:

Enter the organization name: IEEE
Enter a member ID: 1001
Enter a member name: Ahmed
Enter a member age: 31
Total Number of Years: 8
Total Published Papers: 6

Do you want to add another Member? (Yes\No): Yes
Enter a member ID: 1002
Enter a member name: Khalid
Enter a member age: 40
Total Number of Years: 10
Total Published Papers: 9

Do you want to add another Member? (Yes\No): Yes
Enter a member ID: 1003
Enter a member name: Ibrahim
Enter a member age: 41
Total Number of Years: 4
Total Published Papers: 3

Do you want to add another Member? (Yes\No): Yes
Enter a member ID: 1004
Enter a member name: Mohmmed
Enter a member age: 45
Total Number of Years: 5
Total Published Papers: 5

Do you want to add another Member? (Yes\No): Yes

Enter a member ID: 1005

Enter a member name: Sami

Enter a member age: 50

Total Number of Years: 7

Total Published Papers: 2

Do you want to add another Pilot? (Yes\No): No

Organization Name: IEEE

-----Member of the organization-----

Member {ID=1001, name=Ahmed, age=31}

The Record {total Numbers of Years=8, total Published Papers =6}

Member {ID=1002, name=Khalid, age=40}

The Record {total Numbers of Years=10, total Published Papers =9}

Member {ID=1003, name=Ibrahim, age=41}

The Record {total Numbers of Years=4, total Published Papers=3}

Member {ID=1004, name= Mohmmmed, age=45}

The Record {total Numbers of Years=5, total Published Papers=5}

Member {ID=1005, name= Sami, age=50}

The Record {total Numbers of Years=7, total Published Papers=2}

-----List All Members with Reward-----

Member {ID=1001, name=Ahmed, age=31}

The Record {total Numbers of Years=8, total Published Papers =6}

Reward: Discount 50%

Member {ID=1002, name=Khalid, age=40}

The Record {total Numbers of Years=10, total Published Papers =9}

Reward: Discount 50%

Good Luck 😊